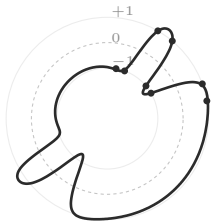


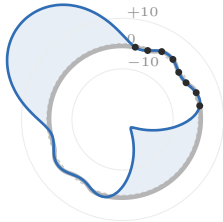
Quadrant, paired labels

Dense network
reference predictor



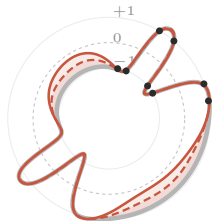
reference

Frozen NTK
initial kernel, all blocks



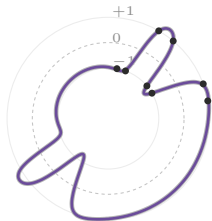
RMS 13.69
output scale $\times 10$

Trained factors
 $W_0 + AB$, rank 24



RMS 0.3755 / 0.2036
two seeds: solid / dashed

Response memory
 $P = 3$, rank bound 24



RMS 0.002818

— dense reference

• training labels

Same initialized network; each model at training MSE $\simeq 0.001$, at its own fitting time.
RMS differences from dense use 8192 circle queries. Signed ticks give output values; any wider NTK scale is marked explicitly.